

CHURNGUARD AI

Customer Churn Prediction & Retention Intelligence System

MainCrafts Technology — 2-Day Skill Certification Program

Domain: Artificial Intelligence & Machine Learning

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1. Problem Statement

Businesses lose customers for many reasons, and companies often discover a customer is leaving only after the decision has already been made. CHURNGUARD AI addresses this by analyzing historical customer data to predict, in advance, whether a customer is likely to churn — flagging a Customer Churn Risk of HIGH, MEDIUM or LOW so the business can intervene before the customer actually leaves.

2. Objective

- Analyze customer data and identify patterns linked to churn.
- Prepare the data for machine learning (cleaning, encoding, splitting).
- Train and compare at least two classification models.
- Evaluate the selected model with appropriate metrics.
- Predict churn risk for new customers, with a probability and a risk level.
- Identify the features that most influence the prediction.
- Translate the results into practical business recommendations.

3. Dataset Description

The project uses a customer-churn dataset structured like the IBM Telco Customer Churn dataset (no internet access was available in the working environment to download the original file directly, so a statistically realistic synthetic dataset with the same columns and churn drivers was generated for the exercise). It contains **7058** customer records and **21** columns, covering demographics (gender, senior citizen, partner, dependents), account information (tenure, contract type, payment method, billing), subscribed services (phone, internet, security, backup, streaming), monthly/total charges, and the target column **Churn** (Yes/No).

4. Technologies Used

- Python 3 — core language
- Pandas / NumPy — data handling
- Matplotlib / Seaborn — visualization
- Scikit-learn — preprocessing (OneHotEncoder, ColumnTransformer, Pipeline), Logistic Regression, Random Forest, evaluation metrics
- Joblib — model persistence
- Google Colab — recommended execution environment

5. Methodology

The project follows the required end-to-end workflow: **Raw Data** → **Data Understanding** → **Cleaning** → **Feature Preparation** → **Model Training** → **Evaluation** → **Prediction** → **Business Insight**.

6. Data Preprocessing

- **TotalCharges** was loaded as text and converted to numeric with `pd.to_numeric(errors='coerce')`, revealing 115 missing values — almost all belonging to brand-new customers with zero tenure. These were filled with `MonthlyCharges × tenure` rather than dropped, since deleting them would discard real customers.
- **15 duplicate rows** were removed with `drop_duplicates()`.
- **customerID** was dropped as a non-predictive unique identifier.
- Categorical columns were one-hot encoded with `OneHotEncoder(handle_unknown='ignore')` inside a `ColumnTransformer`, wrapped in a scikit-learn `Pipeline` so preprocessing and modeling travel together.
- Data was split 80/20 into train (5634 rows) and test (1409 rows) sets, stratified on the target to preserve the churn/non-churn ratio in both sets.

7. Exploratory Data Analysis

Four visualizations were produced to understand churn drivers before modeling.

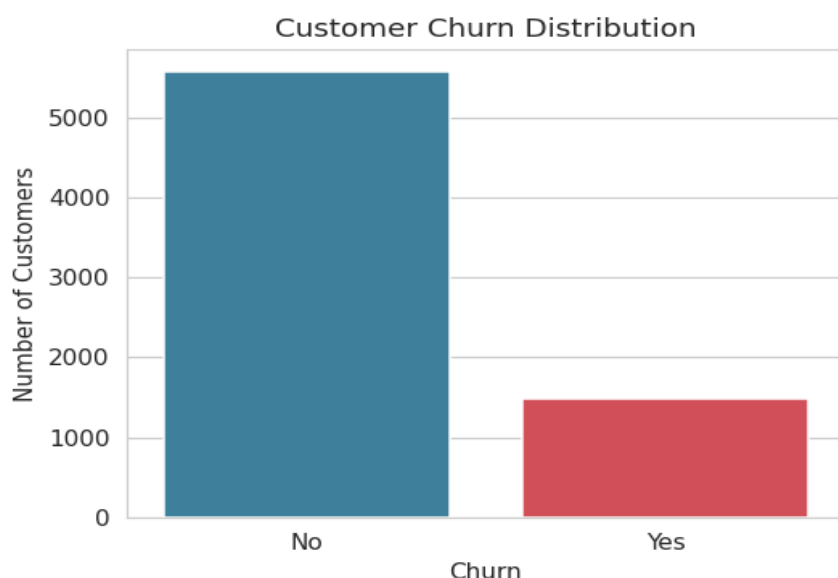


Figure 1 — Churn Distribution. ~20.9% of customers churned; the classes are imbalanced (~4:1), which is why accuracy alone is a misleading metric.

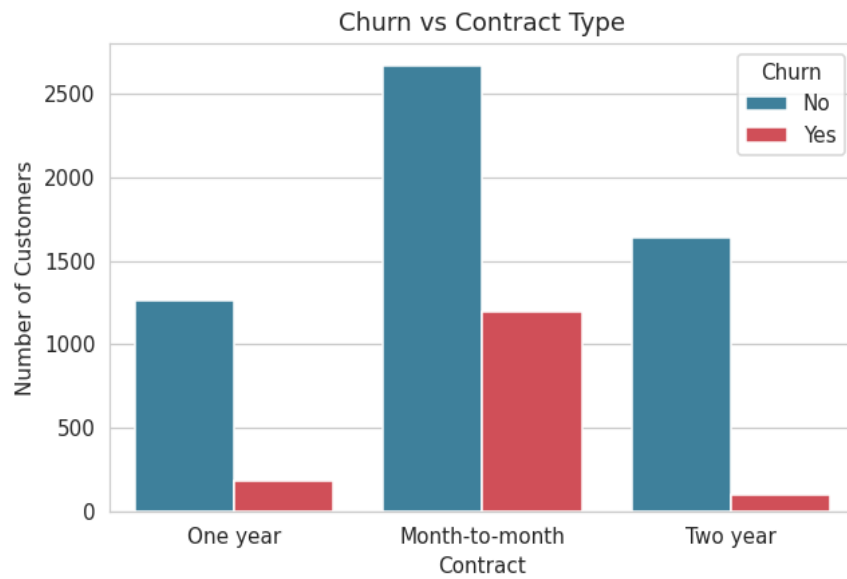


Figure 2 — Churn vs Contract Type. Month-to-month customers churn far more than one/two-year contract holders.

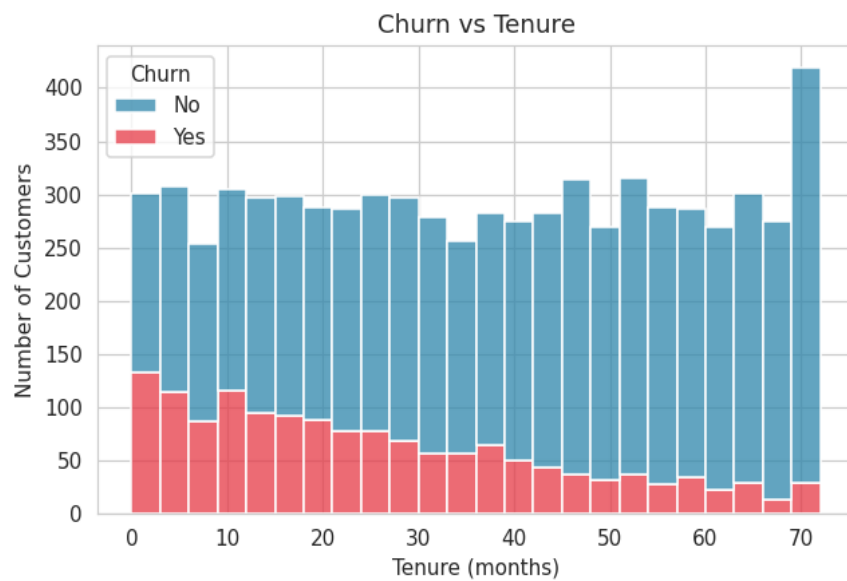


Figure 3 — Churn vs Tenure. Churn concentrates in the first ~12 months; long-tenure customers rarely leave.

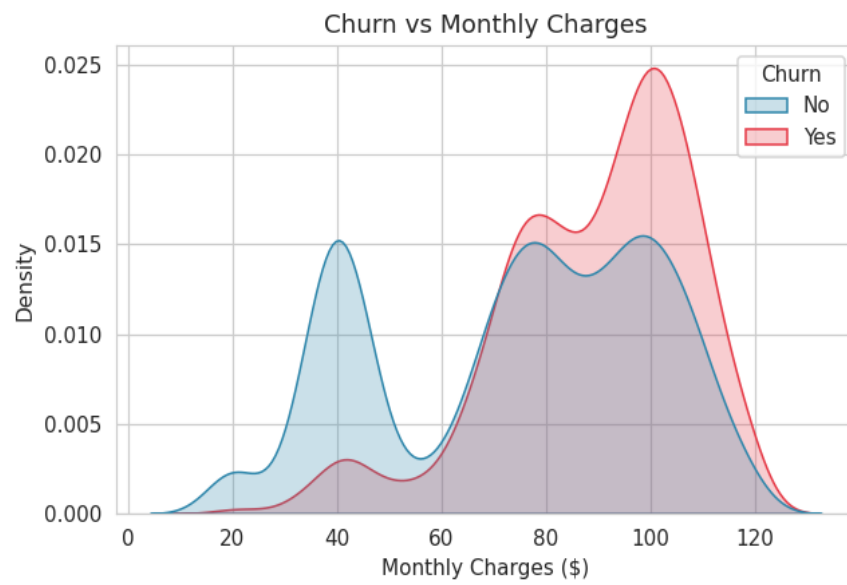


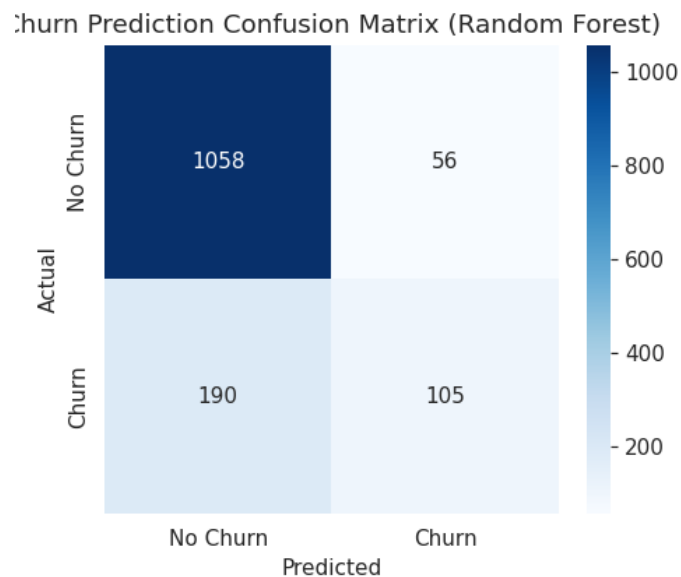
Figure 4 — Churn vs Monthly Charges. Churners skew toward higher monthly charges, consistent with fiber-optic pricing.

8. Models Tested & Evaluation Results

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8261	0.6488	0.3695	0.4708
Random Forest	0.8254	0.6522	0.3559	0.4605

Accuracy alone was rejected as the selection metric given the ~4:1 class imbalance — a model predicting "No churn" for everyone would already score close to 79% while catching no churners. **Random Forest was selected** for further tuning, and recall on the churn class was prioritized because a missed churner (false negative) is a permanent, recurring loss, while a false positive only costs an unnecessary retention offer.

Confusion Matrix (baseline Random Forest)



True Negatives: 1058 · False Positives: 56 · False Negatives: 190 · True Positives: 105. The 190 false negatives — churners the baseline model missed — motivated the Day-2 improvement below.

9. Model Improvement (Day 2, Challenge A)

Before: F1 Score = 0.4605 (Random Forest, default settings).

After: F1 Score = 0.5784 (Random Forest, tuned).

What changed: `class_weight='balanced'` was added to counter the class imbalance, `max_depth=10` and `min_samples_leaf=3` were added to reduce overfitting, and the ensemble was widened to 400 trees.

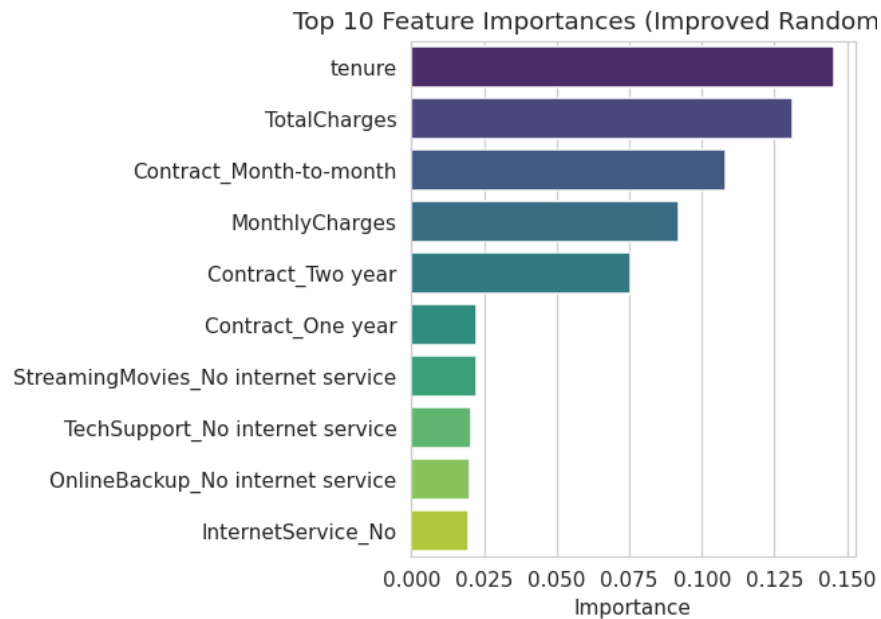
Why: the baseline model under-caught churners (recall 0.36). Class-weighting directly penalizes missed churners more, raising recall to **0.67** and F1 to **0.5784**, at some cost to precision — an acceptable trade-off for a business that would rather over-flag a few loyal customers than miss real churners.

10. Final Model

Metric	Value
Accuracy	0.781
Precision	0.4840

Recall	0.7186
F1 Score	0.5784

Top Feature Importances



The most influential features were **tenure**, **TotalCharges**, **Contract_Month-to-month** — consistent with the EDA findings that tenure, spend and contract type are the strongest churn signals.

Sample Prediction Output

Prediction: **Customer likely to churn** · Churn Probability: **88.0%** · Risk Level: **HIGH** (new month-to-month, fiber-optic customer, 5 months' tenure).

11. Business Insights

- Contract type is the strongest churn lever — moving month-to-month customers to annual contracts is likely the single highest-leverage retention action available.
- Churn risk is front-loaded into the first year; new customers deserve proactive onboarding and check-ins.
- Fiber-optic customers without security/tech-support add-ons, paying by electronic check, form a consistently higher-risk segment worth bundling protective services for.

12. Limitations

- The model outputs a probability, not a certainty — some flagged customers will stay, and some unflagged customers will still leave.
- Improving recall reduced precision; more customers will be flagged than will actually churn, so retention actions should be proportional to the risk tier rather than treated as guaranteed losses.
- Trained on synthetic data mirroring known Telco churn patterns; results on a real production dataset should be re-validated before deployment.

13. Future Improvements

- Test gradient-boosted models (XGBoost/LightGBM) for a possible further F1 gain.
- Add a cost-sensitive threshold (rather than 0.40/0.70) calibrated to actual retention-offer cost vs. customer lifetime value.
- Build the optional Streamlit interface for a live, interactive risk lookup.
- Retrain periodically as new customer data accumulates, to catch drift in churn drivers.